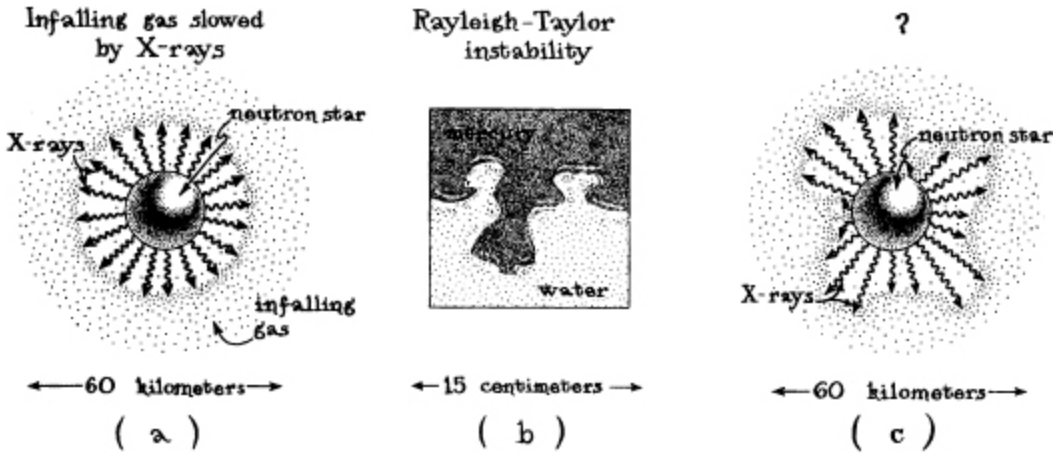


6.4 (a) Gas falling onto a neutron star is slowed by the pressure of outpouring X-rays. (b) Liquid mercury trying to fall in the Earth's gravitational field is held back by water beneath it; a Rayleigh–Taylor instability results. (c) Is it possible that there is also a Rayleigh–Taylor instability for the infalling gas held back by a neutron star's X-rays?

"That's very peculiar," I told Stirling. "Zel'dovich told me precisely the same thing—the flow is stable. But he, like you, would not point me to any proofs." "Oh! That's fascinating. So Zel'dovich really knew," said Stirling.

And then I knew as well. I hadn't wanted to know. But the conclusion was unavoidable. The Teller–Ulam idea must be the use of X-rays, emitted in the first microsecond of the fission (atomic bomb) trigger, to heat, help compress, and ignite the superbomb's fusion fuel (Figure 6.5). That this is, indeed, part of the Teller–Ulam idea was confirmed in the 1980s in several unclassified American publications; otherwise I would not mention it here.

